Final

Will Grimsich

2026-06-09

Table of contents

Github Repository **2**

Set Up **2**

Problems **2**
Problem 1. Research Writing . . . 2
a. Transparent Statistical Methods . . . 2
b. Suggestions for Rewriting . . . 2
c. Further Analysis . . . 3
Problem 2. Data Analysis . . . 4
a. Clean & Wrangle . . . 4
b. Response Variable . . . 4
c. Purpose of Study . . . 4
d. Table of Models . . . 4
e. Running the Models . . . 4
f. Selecting the Best Model . . . 4
g. Checking Diagnostics . . . 4
h. Visualizing Model Predictions . . . 4
i. Writing a Caption for Figure . . . 4
j. Calculating Model Predictions . . . 4
k. Interpreting Results . . . 4
Problem 3. Affective & Exploratory Visualizations . . . 4
a. Comparing Visualizations . . . 4
b. Designing an Analysis . . . 4
c. Checking Assumptions & Running Analysis . . . 4
d. Creating Visualization . . . 4
e. Writing Caption . . . 4
f. Writing About Results . . . 4
g. Sharing Affective Visualization . . . 4

Github Repository

Set Up

```
library(tidyverse)
library(here)
library(janitor)

nest_data <- read.csv(here("data","nest_data_final.csv"))
```

Problems

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, they used a binomial "log odds" regression to relate and predict (*Recurvirostra americana*) habitat occupation to distance from water edge.

In part 2, they used a Chi-Squared test to compare expected vs observed counts for habitat type and occupation by species.

b. Suggestions for Rewriting

Utilizing binomial logistic regression analysis, we found a significant relationship between distance from the water's edge and American Avocet (*Recurvirostra americana*) microhabitat use.

Utilizing a Chi-Squared test, we found a significant relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further Analysis

One additional variable that could influence the probability of American Avocet (*Recurvirostra americana*) microhabitat use is **salinity.** This variable is continuous, numerical (ppm) and often calculated after measuring electrical conductivity in micro Siemens per centimeter (µS/cm). Salinity influences a plethora of biological interactions, specifically dehydration, and decreases in weight, feeding and preening in American Avocet chicks.

Problem 2. Data Analysis

a. Clean & Wrangle

b. Response Variable

c. Purpose of Study

d. Table of Models

e. Running the Models

f. Selecting the Best Model

g. Checking Diagnostics

h. Visualizing Model Predictions

i. Writing a Caption for Figure

j. Calculating Model Predictions

k. Interpreting Results

Problem 3. Affective & Exploratory Visualizations

a. Comparing Visualizations

b. Designing an Analysis

c. Checking Assumptions & Running Analysis

d. Creating Visualization

e. Writing Caption

f. Writing About Results

g. Sharing Affective Visualization